Linear Programming

Duality:

The primal is given by

$$\text{Max. } Z = 40x_1 + 35x_2 \text{ subject to}$$

The Problem is called PRIMAL

$$2x_1 + 3x_2 \leq 60$$

$$4x_1 + 3x_2 \leq 96$$

$$x_1, x_2 \geq 0$$

Find duality.

Conditions for Duality:

for Maximization, all constraints $\leq$ RHS

Minimization, all constraints $\geq$ RHS

Dual is given by

$$\text{Minimize. } W = 60y_1 + 96y_2 \text{ subject to}$$

$$2y_1 + 4y_2 \geq 40$$

$$3y_1 + 3y_2 \geq 35$$

$$y_1, y_2 \geq 0$$

Find the duality of

$$\text{Min. } Z = 40x_1 + 20x_2 \text{ subject to}$$

PRIMAL

$$3x_1 + 2x_2 \geq 48$$

$$x_1 + 3x_2 \geq 81$$

$$2x_1 - x_2 \leq 61$$

$$x_1, x_2 \geq 0$$

Conditions for duality

For Minimization, all constraints $\geq$ RHS

The primal can be correctly rewritten as

$$\text{Min. } Z = 40x_1 + 20x_2 \text{ subject to}$$

$$3x_1 + 2x_2 \geq 48$$

$$x_1 + 3x_2 \geq 81$$

$$-2x_1 + x_2 \geq -61$$

The dual is given by

$$\text{Max. } W = 48y_1 + 81y_2 - 61y_3 \text{ subject to}$$

$$3y_1 + 1y_2 - 2y_3 \leq 40$$

$$2y_1 + 3y_2 + 1y_3 \leq 20$$

$$y_1, y_2, y_3 \geq 0$$

Find the duality of

Max $Z = x_1 + 2x_2$ subject to

$2x_1 + 4x_2 \leq 160$

$x_1 - x_2 = 30$

$x_1 \geq 10$

$x_1, x_2 \geq 0$

In the constraint 2, there is equality sign.

Equality can be $\geq$ or $\leq$

Since the question is of Maximization type

As a rule, we use constraints $\leq$ RHS for maximization

Here, we have 2 symbols $\geq$ and $\leq$.

we first use $\leq$ (since Maximization problem)

subsequently we use $\geq$ in the constraint 2 above.

Max. $Z = x_1 + 2x_2$ subject to

$2x_1 + 4x_2 \leq 160$

$x_1 - x_2 \leq 30$

$x_1 - x_2 \geq 30$

$x_1 \geq 10$

$x_1, x_2 \geq 0$

Remember

Max → RHS max

① $\leq$

② $\geq$

we rewrite the above equation for Maximization using symbol $\leq$ for all the constraints

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Max. $Z = 1x_1 + 2x_2$ subject to

y_1 $\quad 2x_1 + 4x_2 \leq 160$

y_2 $\quad 1x_1 - 1x_2 \leq 30$

y_3 $\quad -1x_1 + 1x_2 \leq -30$

y_4 $\quad -1x_1 \leq -10$

$x_1, x_2 \geq 0$

For dual, we minimize the function

$$\text{Min } W = 160y_1 + 30y_2 - 30y_3 - 10y_4$$

subject to

We change symbol to $\geq$ for Min.

$$2y_1 + 1y_2 - 1y_3 - 1y_4 \geq 1$$

$$4y_1 - 1y_2 + 1y_3 + 0y_4 \geq 2$$

$$y_1, y_2, y_3, y_4 \geq 0$$

$$\text{Min } W = 160y_1 + 30y_2 - 30y_3 - 10y_4$$

Since we split the equal to into 2 parts, we merge the 2 together

$$\text{Min } W = 160y_1 + 30(y_2 - y_3) - 10y_4$$

Let $y_2 - y_3 = y'$ (Replace $y_2 - y_3$ with y')

We rewrite the objective function and the constraints as

Min $W = 160y_1 + 30y' - 10y_4$ subject to

$2y_1 + y' - y_4 \geq 1$

$4y_1 - y' \geq 2$

$y_1, y_4 \geq 0$

y' is unrestricted $y' \geq 0$ or $y' \leq 0$